



Anglo-Chinese Junior College

JC2 Biology Preliminary Examination

Higher 2



A Methodist Institution
(Founded 1886)

CANDIDATE
NAME

FORM
CLASS

TUTORIAL
CLASS

INDEX
NUMBER

BIOLOGY

Paper 2 Structured Questions

9744/02

25 August 2022

2 hours

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Name, Class and Index number in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiners' use only	
1	/ 9
2	/ 10
3	/ 11
4	/ 9
5	/ 10
6	/ 10
7	/ 10
8	/ 10
9	/ 10
10	/ 6
11	/ 5
Total	/ 100

Answer **all** questions.

- 1 Fig. 1.1 shows *Mycobacterium tuberculosis* infecting an alveolar cell in the lungs.

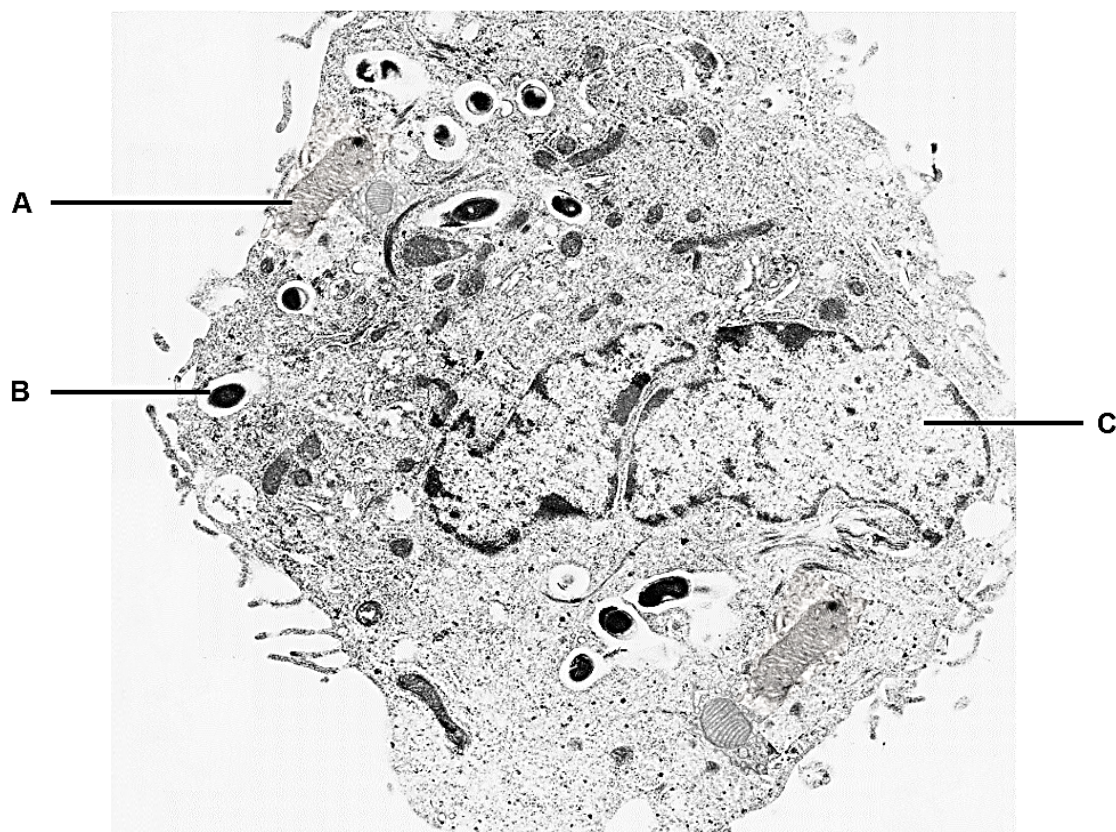


Fig. 1.1

- (a) (i) Using the letters **A** to **C**, identify the mitochondrion.

A;

..... [1]

- (ii) Justify your answer to (a)(i).

1. **Double membrane-bound**;
2. **Highly folded inner membrane**;
3. **Rod-shaped**;

max 2m

..... [2]

- (b) Compare the structures of *M. tuberculosis* and a mitochondrion.

Similarity:

1. Both contain circular DNA;
2. Both contain 70S ribosomes;

Difference:

3. *M. tuberculosis* / bacterial cells is bound by a **single** membrane, while a mitochondrion is bound by a **double** membrane;
4. *M. tuberculosis* has a peptidoglycan cell wall, while mitochondrion does not have a cell wall;

At least one similarity and one difference

..... [3]

(c) Explain how the structure of organelle **C** allows it to serve its functions.

1. Presence of a nuclear envelope / bound by double membrane, which allows nucleus to **protect the DNA / chromatin from reactions in the cytoplasm**;
2. Presence of nuclear pores which allow the **export of mRNA/ribosomal subunits for translation in the cytoplasm / import of ribosomal proteins for assembly of ribosomal subunits / import of nucleotides for DNA replication or transcription**;
3. Presence of enzymes in the nucleoplasm which allows DNA replication / transcription / **post-transcriptional modification** to occur;
4. Presence of nucleolus which allows the **synthesis of rRNA / assembly of ribosomal subunits**;

max 3m

..... [3]

[Total: 9]

[Turn over

- 2 The rate of carbon dioxide uptake at a range of carbon dioxide concentrations by barley and sugar cane were compared at two temperatures using the apparatus shown in Fig. 2.1.

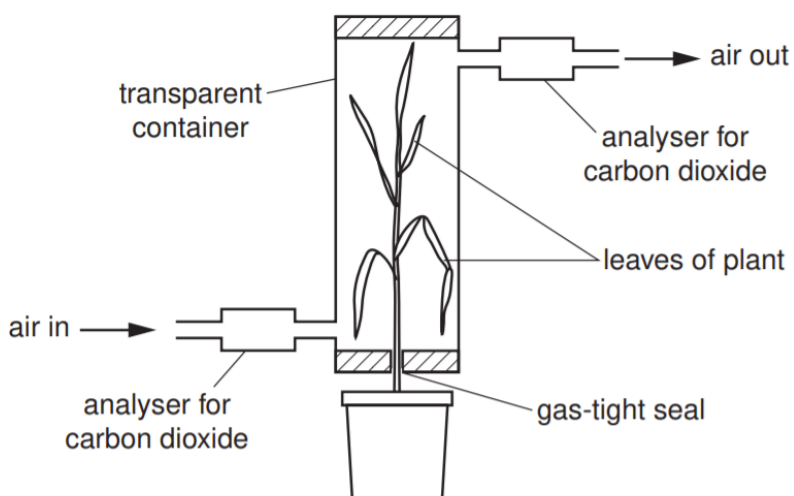


Fig. 2.1

The results of the experiment are presented in Fig. 2.2.

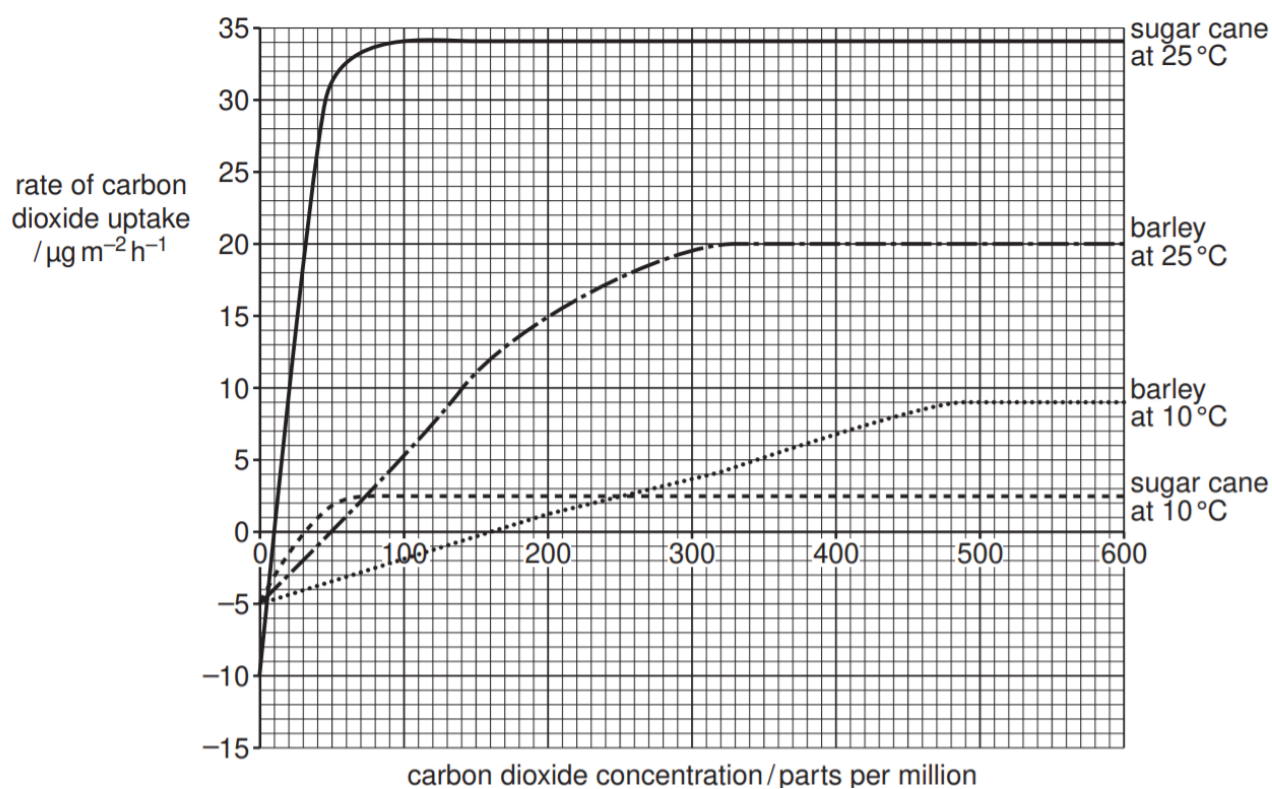


Fig. 2.2

- (a) (i) With reference to Fig. 2.2, describe the differences in rates of carbon dioxide uptake by barley and sugar cane in response to increasing carbon dioxide concentrations at 25°C.
1. **Sugar cane** reaches **maximum rate** of CO₂ uptake at **lower** CO₂ concentration / **barley** reaches **maximum rate** of CO₂ uptake at **higher** CO₂ concentration;
 2. Quote data: maximum rate for sugar cane is reached at 100ppm vs barley at 320ppm at 25°C;
 3. Sugar cane reaches a **higher maximum rate** of carbon dioxide uptake than barley;
 4. Quote data: 34 $\mu\text{g m}^{-2} \text{h}^{-1}$ for sugar cane vs 20 $\mu\text{g m}^{-2} \text{h}^{-1}$ for barley;
 5. The **rate of CO₂ uptake** by sugar cane **increases faster** than the rate of CO₂ uptake by barley as CO₂ concentration increases;
 6. Rate of CO₂ uptake by sugar cane increases faster from 0 to 34 $\mu\text{g m}^{-2} \text{h}^{-1}$ compared to that by barley from 0 to 5.5 $\mu\text{g m}^{-2} \text{h}^{-1}$ as CO₂ concentration increases from 0ppm to 100ppm;
 7. Sugar cane has a **lower** CO₂ compensation point than barley;
 8. CO₂ compensation point for sugar cane is 10ppm vs. barley at 50ppm at 25°C;
2 trends + corresponding data, max 4m
- [4]
- (ii) Barley and sugar cane thrive in different climatic conditions. Complete Table 2.1 to show where these plants are likely to be found.

Table 2.1

climatic conditions	plant
tropical, warm climate	sugar cane
temperate, cool climate	barley;

[1]

[Turn over]

Sugar cane is a cash crop which produces a high concentration of sucrose. Invertase found in the sugar cane hydrolyses sucrose into glucose and fructose, resulting in post-harvest loss of sucrose.

(b) Explain why a non-competitive inhibitor of invertase can potentially be used to reduce such losses.

1. A non-competitive inhibitor **binds to a site other than the active site** of invertase;
2. **change the 3D conformation** of invertase such that it is **no longer complementary to sucrose** **does not bind to sucrose/ sucrose hydrolysis is prevented**;

..... [2]

Starch is also found in sugar cane as a storage polysaccharide.

(c) (i) Describe the advantages to a plant of condensing glucose molecules into starch.

1. starch is **more compact**, making it a better energy **storage** molecule;
2. starch is **insoluble in water**/ osmotically inactive or inert/ does not affect osmotic potential in cells, thus a better **storage** molecule;

function as "storage molecule" to be stated at least once in MP1 or MP2

..... [2]

(ii) Suggest why mammals store glucose as glycogen rather than as starch.

1. glycogen has **more branching points** than amylopectin in starch, which allow for **fast mobilisation of glucose** when needed/ can be **compacted further for energy storage** purposes;
2. **absence of enzymes** in mammals that can **polymerise** glucose to form starch;

max 1m

..... [1]

[Total: 10]

- 3 Table 3.1 shows the telomerase activity and change in telomere length over time in different human stem cells.

Table 3.1

type of stem cell	telomerase activity	telomere length over time
germline	high	maintained
zygotic	high	maintained
embryonic	high	maintained
blood	low	shortened
skin	low	shortened
intestinal	low	shortened

- (a) (i) With the exception of germline stem cells, explain how the potency of stem cells is related to the telomere length in these cells.

1. **Totipotent and pluripotent stem cells exhibit high telomerase activity, while multipotent stem cells exhibit low telomerase activity;**
2. **Telomere lengths are only maintained in cells where telomerase activity is high / will shorten over time in cells where telomerase activity is low;**
3. **With each round of DNA replication, the 5' end of daughter DNA strands is shortened due to the end-replication problem;**
4. **Telomerase synthesises multiple DNA repeats at the 3' end of DNA to lengthen the telomere;**

..... [4]

- (ii) Germline stem cells are the diploid cells that can only give rise to gametes, hence are considered unipotent.

Suggest why telomerase activity is high in germline stem cells.

To maintain the length of telomeres over successive generations of a species / To preserve genetic integrity in the offspring/descendants;

..... [1]

[Turn over

- (iii) Like telomeres, the centromere is an example of a non-coding DNA.

Compare the structures of telomeres and centromeres.

Similarity:

- Both are made of **DNA repeats** / **repetitive sequences of DNA** / **tandem repeats**;

Differences

- Telomere is found at the **ends of** linear **chromosomes** unlike centromere/ while centromeres are found at a point along the chromosomes/ **not at the ends** of chromosomes;
- Telomeres occur in **two regions** on a chromosome while centromere occurs at **one region** on a chromosome;
- Centromeres contain **binding site** complementary to **kinetochore** proteins while telomeres do not have such binding sites;
- Telomeres can form loops while centromeres do not;

At least one similarity and one difference

..... [2]

The Hayflick limit is the number of times a cell population can divide before cell division stops, assuming an absence of telomerase activity.

Healthy human cells start off with an average telomere length of 10 kilobases (kb). Telomeres shorten at rates of 140 to 210 bases per replication prior to each cell division. It is estimated that when telomere lengths reach the threshold of 1.6 kb, the cells become senescent.

- (b) (i) Using the information provided, calculate the range of values for the Hayflick limit.

Lower range of the Hayflick limit = $(10,000 - 1600) / 210 = 40$ divisions

Upper range of the Hayflick limit = $(10,000 - 1600) / 140 = 60$ divisions

- Complete workings shown – 1m
- Correct answer – 1m

..... to divisions [2]

- (ii) Describe two possible deleterious effects on DNA when telomeres are depleted.

- Genes** near / at the ends of DNA may **become eroded**;
- DNA ends / chromosomal ends** may undergo **end-to-end fusions**;
- DNA ends / chromosomal ends** may be **degraded by exonucleases**;

R! replicative senescence and apoptosis

max 2m

..... [2]

[Total: 11]

- 4 One of the three tenets of the cell theory states that all living organisms are composed of one or more cells. Viruses challenge this tenet as they possess both living and non-living characteristics.

Giant viruses such as mimiviruses have been discovered recently. It is debatable whether such viruses are considered as living organisms or non-living entities. Mimivirus contains a DNA genome which is significantly larger than the genomes of any other known virus and is comparable to that of a cell. Like other viruses, it is dependent on its host for translation.

Fig. 4.1 shows the structure of Mimivirus.

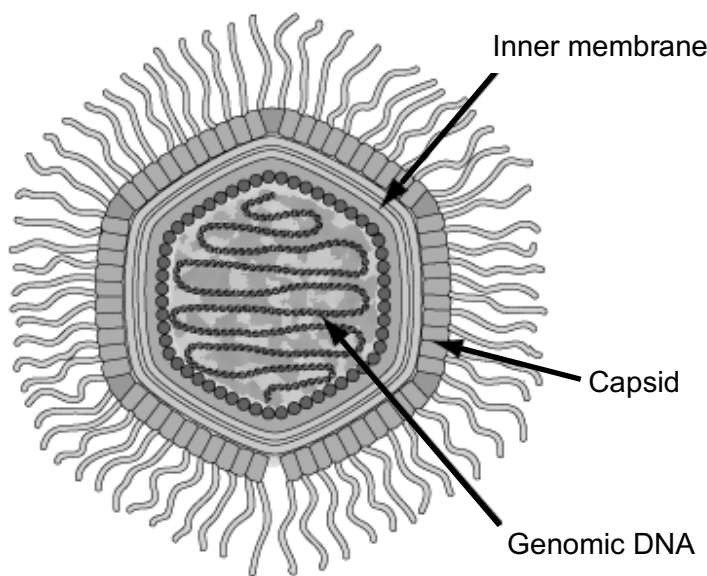


Fig. 4.1

- (a) With reference to the information provided, discuss if the discovery of mimivirus justify the classification of viruses as living.

Classify as living organisms:

1. It has genes/ genetic material which **codes for proteins**;

Classify as not living organisms:

2. It does **not have ribosomes/ cannot synthesise proteins to replicate itself independently of the host cell/ still require host to reproduce/ not free-living**;
3. According to Fig. 4.1, it **does not have organelles**;

..... [3]

[Turn over

Fig. 4.2 shows the structure of an influenza virus.

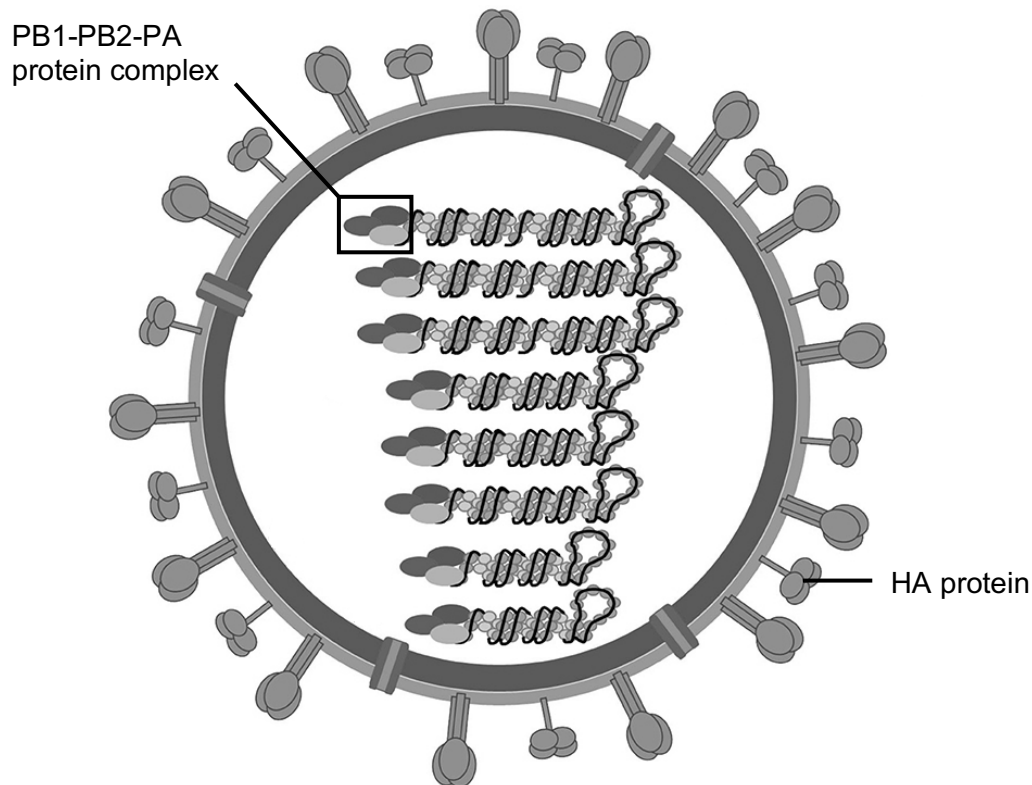


Fig. 4.2

(b) Describe the role of the following components labelled in Fig. 4.2 in the viral reproductive cycle:

(i) PB1-PB2-PA protein complex.

1. PB1, PB2 and PA form the viral RNA-dependent RNA polymerase/ RNA replicase;
2. which transcribes the (-)-sense RNA genome to form complementary (+)-sense mRNA;
3. These (+)-sense mRNA can be translated to form viral proteins/ used as templates to make new copies of the (-)-sense RNA genome;

max 2m

..... [2]

(ii) HA protein.

1. HA protein is haemagglutinin which binds to specific sialic acid receptors on the cell surface of the epithelial cells lining the respiratory tract;
2. allows virus to be taken into cells via receptor-mediated endocytosis;

..... [2]

Oseltamivir is an orally administered antiviral medication that is recommended for patients infected with influenza virus. Fig. 4.3a shows the mechanism of action of oseltamivir and Fig. 4.3b shows its molecular structure as compared to that of sialic acid.

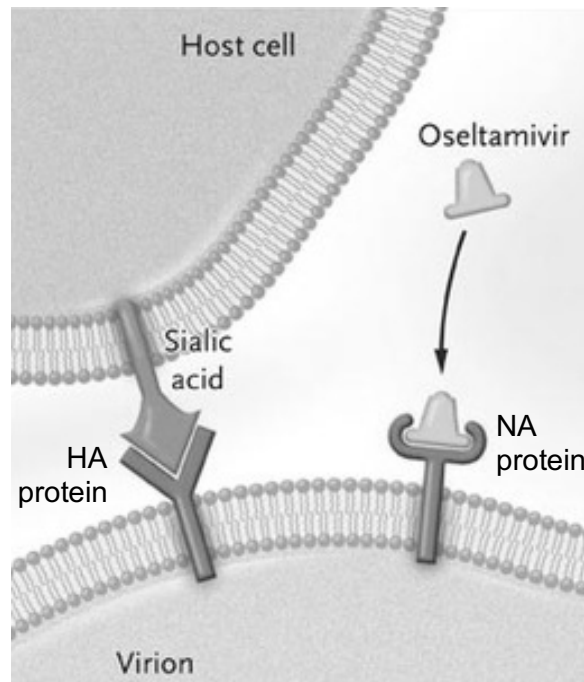


Fig. 4.3a

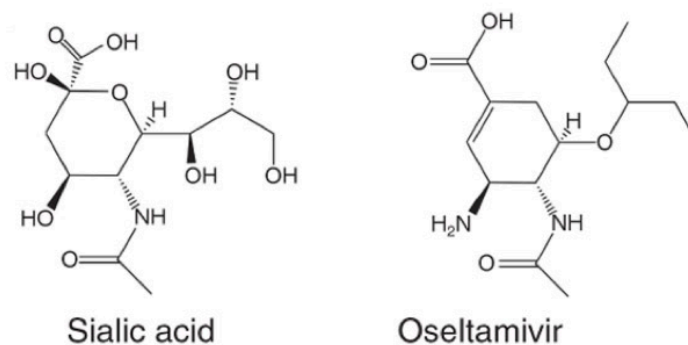


Fig. 4.3b

(c) Explain how oseltamivir affects the reproductive cycle of H1N1 virus.

1. Oseltamivir has **similar 3D conformation** as sialic acid and is able to bind to the **active site of neuraminidase**;
2. **Prevents neuraminidase from cleaving sialic acid** so that virion **cannot be released via budding** to infect other cells;

[2]

[Total: 9]

[Turn over

5 Fig. 5.1 shows the expression of the *lac* operon.

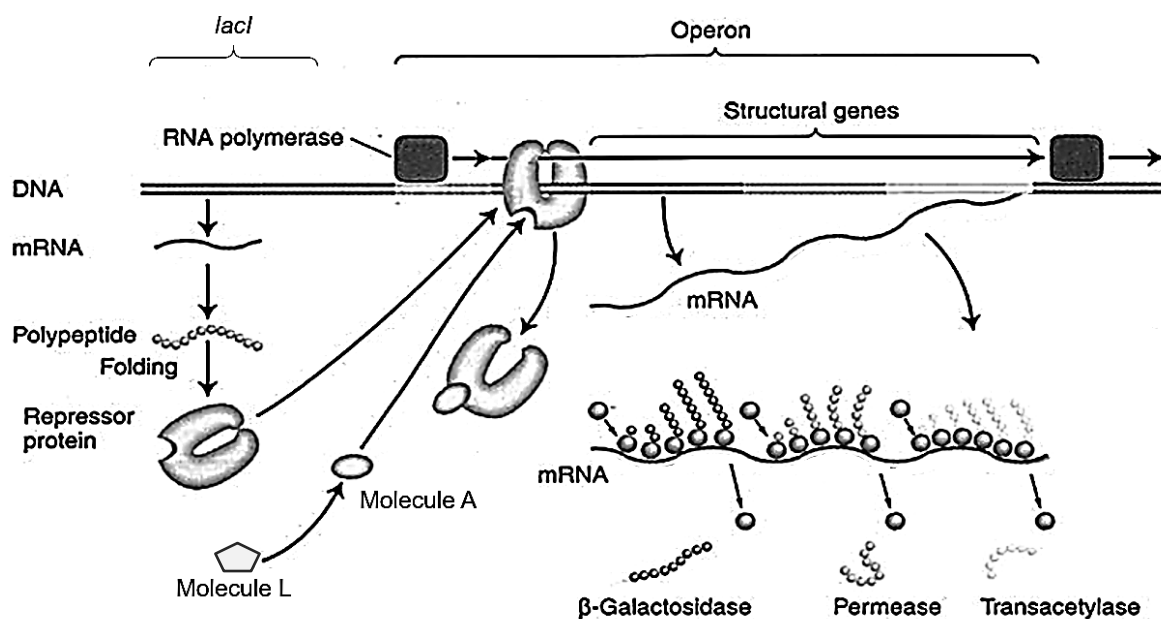


Fig. 5.1

(a) With reference to Fig. 5.1, identify molecule A.

Allolactose; R! lactose

[1]

An investigation was conducted to find out more about the *lac* operon. Bacteria were grown in a medium containing 2% glucose and 2% lactose. Three different experiments were conducted.

- experiment **A** – Wild type bacteria.
- experiment **B** – Bacteria with a mutation.
- experiment **C** – Wild type bacteria with IPTG added. IPTG is a molecular mimic of allolactose.

Fig. 5.2 shows the results of the investigation.

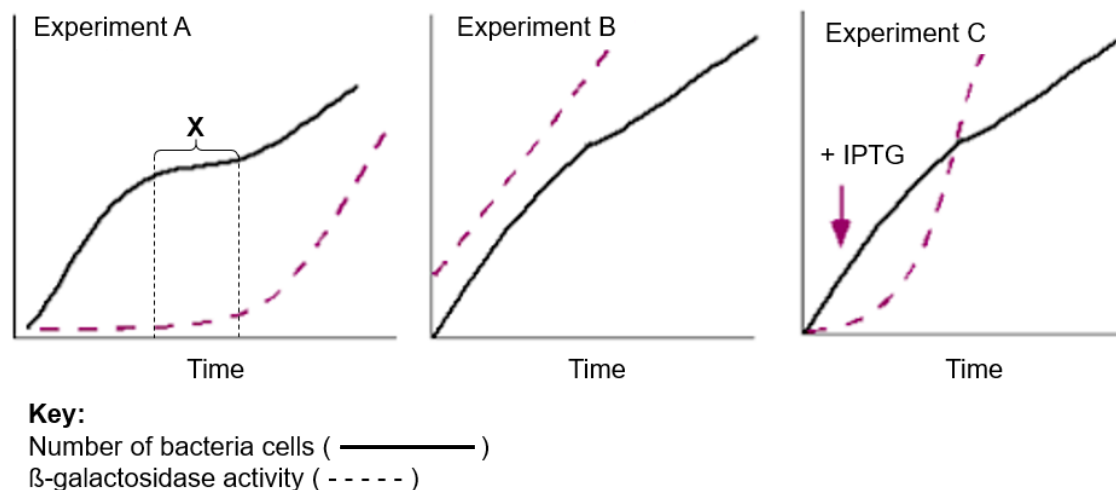


Fig. 5.2

- (b) (i) Explain the bacterial growth during period **X** in experiment **A**.
1. *Number of bacteria **remains constant** as **glucose** has been **used up**;
 2. **Time is needed** for the activation of **adenyl cyclase** at low glucose concentration to increase **cAMP** level;
 3. cAMP then binds and activate **CAP** to stimulate the **transcription of the lac operon to produce more β -galactosidase** to metabolise more lactose;
- *MP1 compulsory, max 2m
- [2]
- (ii) Suggest what mutation might have occurred in the bacteria used in experiment **B**.
1. Loss of function mutation in **lacI gene** coding for lac repressor;
 2. Resulting in the production of **non-functional/ inactive lac repressor, unable to bind to operator**;
 3. Mutation of **operator** sequence of lac operon;
 4. Resulting in active **lac repressor unable to bind** to operator;
- *MP1 + MP2 OR MP3 + MP4
- [2]
- (iii) Explain the effect of IPTG in experiment **C**.
1. Acts as inducer which binds to allosteric site of active **lac repressor** to **inactivate** it;
 2. repressor **unable to bind** to **operator**, resulting in transcription of lac operon/ β -galactosidase gene;
- [2]

[Turn over]

(c) Explain how bacteriophages could enable more bacteria in a population to metabolise lactose.

1. In generalised transduction, a random segment/ fragment of the infected bacteria cell's degraded DNA containing the *lacZ* gene / *lac* operon* (R! bacterial chromosome/ genome) is mistakenly packaged into a phage;

OR

In specialised transduction, when prophage leaves bacterial chromosome, prophage/ viral DNA may be improperly excised, a specific segment/ fragment of bacterial DNA containing the *lacZ* gene / *lac* operon* that is adjacent to prophage is excised with it;

2. Upon its release from the lysed host/ bacteria cell, the phage can infect/ attach to another bacterium (recipient) and inject/ transfer the piece of bacterial DNA (R! bacterial chromosome/ genome, viral genome/ DNA) acquired from the donor;
3. DNA of donor cell can subsequently replace homologous region of chromosome of recipient cell via homologous recombination (for both generalised and specialised transduction)/ incorporated into host chromosome via integration (for specialised transduction);

*** Must have mention of *lacZ* / *lac* operon in answer**

..... [3]

[Total: 10]

- 6 Cancer is a disease in which genetic alterations, most frequently chromosomal aberrations, are acquired in somatic cells. Fig. 6.1 shows a karyotype from a patient diagnosed with leukemia.



Fig. 6.1

- (a) (i) Describe two types of chromosomal aberrations evident in Fig. 6.1.

1. Aneuploidy resulting from the loss of one chromosome 7;
2. Translocation of a segment of chromosome 4 to chromosome 11;

..... [2]

[Turn over

- (ii) Acquired chromosome aberrations affecting specific genes are found to be associated with specific types of cancer, such as leukemia.

Explain how the chromosomal aberrations identified in (a)(i) could have resulted in uncontrolled division of cells in a leukemia patient.

1. Aneuploidy may have led to a **loss of tumour suppressor genes** / translocation may have led to a **disruption of tumour suppressor genes**, such that **tumour suppressor proteins are not produced/ are non-functional**;
2. cells with damaged DNA bypass cell cycle checkpoints/ cannot undergo DNA repair/ apoptosis;
3. translocation may have brought a **proto-oncogene under the control of a strong promoter** such that **excess proteins are produced**/ formation of an **oncogene** which codes for a **hyperactive protein**;
4. resulting in cell cycle progression despite a lack of growth signals;

..... [4]

The effect of papaya peel extract on the ability of human tumour cells to divide was investigated. Different concentrations of papaya peel extract were added to cultures of human liver tumour cells, which were then incubated at 37°C. Table 6.1 shows the percentage of tumour cells that are still able to divide after incubation for 48 hours.

Table 6.1

papaya peel extract concentration/ $\mu\text{g cm}^{-3}$	percentage of tumour cells that are able to divide
0	100
5	63
10	58
20	46
40	40
80	37

- (b) (i) Suggest how the papaya peel extract could prevent the tumour cells from dividing by targeting specific phases of mitosis.

Extract may contain enzymes inhibitors/ proteins which	E.g. possible mechanism
1. Inhibit prophase;	2. by preventing chromatin condensation/ nuclear envelope disintegration ;
3. Inhibit metaphase;	4. by preventing spindle fibres attachment to kinetochore complex at the centromere of chromosomes ;
5. Inhibit anaphase;	6. by preventing separation of sister-chromatids to opposite poles of the cell;
7. Inhibit telophase;	8. by preventing formation of nuclear envelope;

Max 1m for named stage + 1m for corresponding mechanism

..... [2]

- (ii) A student stated that $80 \mu\text{g cm}^{-3}$ of papaya peel extract would be suitable as a treatment for skin tumours in humans.

State the reasons why this statement is **not** supported by this study.

1. **37% of cancer cells are still able to divide**;
2. **study is done on liver cancer cells, not on skin cancer cells, so effectiveness of extract on skin cancer cells is unknown**;
3. unknown side effects;
4. **effects *in vivo* are unknown**;
5. skin may **not be permeable to extract** if applied topically/ **unknown concentration arrives at cells** if extract is ingested;

max 2m

..... [2]

[Total: 10]

[Turn over

- 7 Watermelon is a popular fruit crop. The colour intensity of watermelon flesh is a characteristic that some consumers look out for, as there is a perceived positive correlation of flesh colour intensity with sweetness.

Flesh colour intensity is controlled by gene **R/r**, in which allele **R** gives rise to scarlet red flesh (darker shade), while allele **r** gives rise to coral red flesh (lighter shade). Analysing the sequence of the alleles showed that allele **R** had a restriction site for the restriction enzyme *Hpa*II. This restriction site is absent in allele **r** due to a single nucleotide substitution.

Gene **T/t** codes for a sugar transporter which is responsible for different sweetness levels in the different cultivars.

The scientists then carried out an analysis of three generations of watermelon plants (I to III). The fruits produced by each plant were assessed for two features:

- the presence or absence of the *Hpa*II restriction site
- sweetness (the fruits were classified “sweet” or “not as sweet”)

Fig. 7.1 shows the pedigree of the three different generations of watermelon plants.

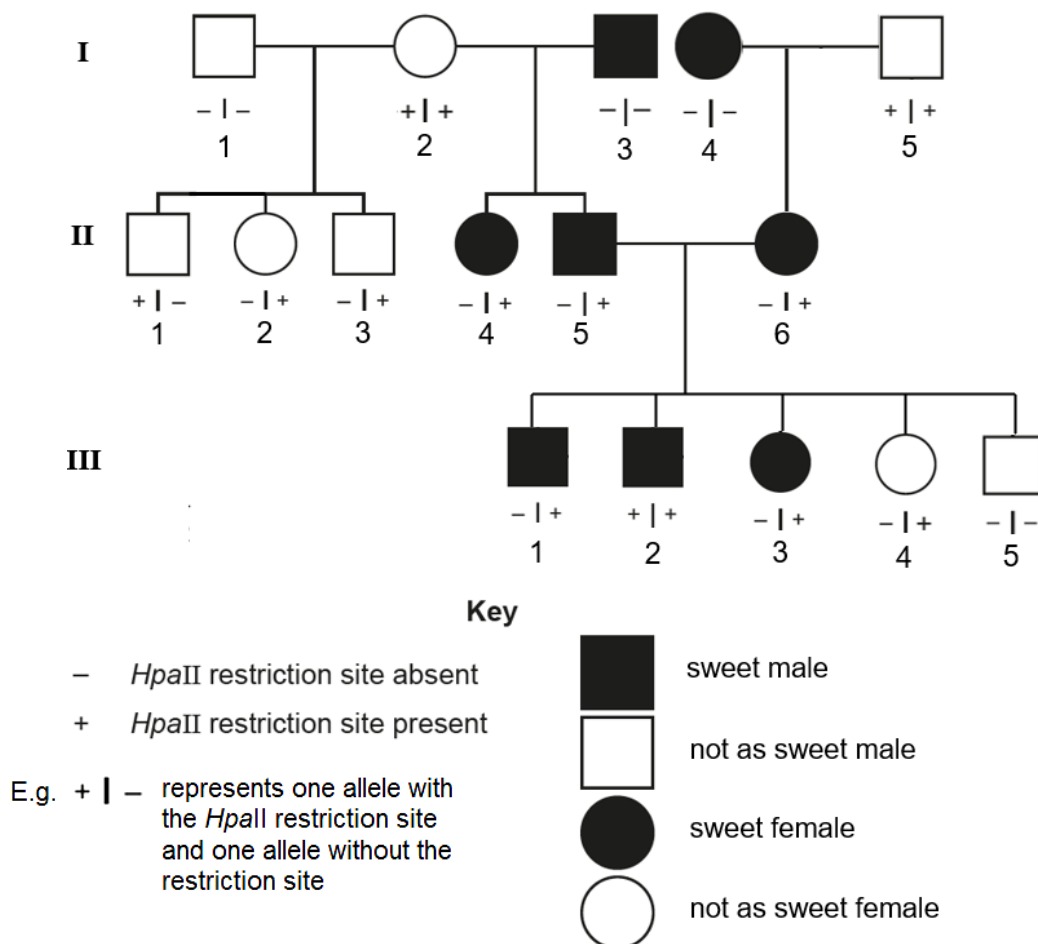


Fig. 7.1

(a) With reference to Fig. 7.1, identify an evidence that supports these statements:

(i) the sweet phenotype is due to a dominant allele.

Individuals II-5 and II-6 are both sweet, but they give rise to offspring which are sweet and not so sweet, hence indicating that II-5 and II-6 are **heterozygotes** (and the not-so-sweet phenotype will only be phenotypically expressed in homozygous recessive individuals);

..... [1]

(ii) the gene **R/r** is located on an autosome.

1. **All individuals** have **two** alleles (which can be seen from having **-/-**, **-/+** or **+/+** in all individuals;

2. **Phenotype** of male offspring is not determined only by the female parent/mother e.g. **III-5 from II-6**, homozygous recessive phenotype due to inheritance from both parents;

max 1m

..... [1]

(b) The perceived positive correlation of flesh colour intensity and sweetness could be due to complete linkage between genes **R/r** and **T/t**. Using the symbols provided, draw a genetic diagram to show the expected phenotypic ratio for the cross between individuals II-5 and II-6 if genes **R/r** and **T/t** are completely linked.

Parental phenotype	Scarlet Red, Sweet x Scarlet Red, Sweet	
Parental genotype	(Rt)(rT) x (Rt)(rT) ;	
Gametes	(Rt) (rT) ; (Rt) (rT) ;	
	(Rt)	(rT)
(Rt)	(Rt)(Rt) Scarlet red, Not so sweet	(Rt)(rT) Scarlet red, Sweet
(rT)	(rT)(Rt) Scarlet red, Sweet	(rT)(rT) Coral red, Sweet

Correct genotype;
Correct phenotype;

Offspring phenotypic ratio

1 Scarlet red, Not so sweet : 2 Scarlet red, Sweet : 1 Coral red, Sweet ;

[5]

[Turn over

- (c) State the expected phenotypic ratio for the cross between individuals II-5 and II-6, if genes **R/r** and **T/t** segregate independently.

9 scarlet red flesh and sweet : 3 scarlet red flesh and not so sweet : 3 coral red flesh and sweet : 1 coral red flesh and not so sweet;

***compulsory to see phenotypes, not just the numbers**

..... [1]

- (d) The scientists aim to verify if there is a positive correlation between flesh colour intensity and sweetness.

A cross is carried out between two individuals which are heterozygous for both the genes **R/r** and **T/t**.

State the null hypothesis and statistical test that would need to be carried out in order to verify the correlation.

Null hypothesis:

The expected ratio of the cross follows a Mendelian inheritance (9:3:3:1), hence the difference between the observed and expected numbers is due to chance / not significant;

Statistical test:

Chi-Squared test;

..... [2]

[Total: 10]

- 8 The Galapagos marine iguanas (*Amblyrhynchus cristatus*) inhabit the coastlines of islands throughout the Galapagos archipelago. They mainly live in colonies on rocky shores, where they bask after foraging for underwater algae and seaweed in the relatively cold waters. The rocks on which they bask on are occasionally exposed to heavy waves. Like other reptiles, they are ectotherms with limited physiological means of keeping their body temperature constant and often rely on external sources of heat.

Morphological and genetic data have revealed that the closest relative of *Amblyrhynchus* is the genus of terrestrial iguanas, *Conolophus*, which is also endemic to the Galapagos.

Fig. 8.1 shows the overall morphology of the marine and terrestrial iguanas, as well as their claw structures.

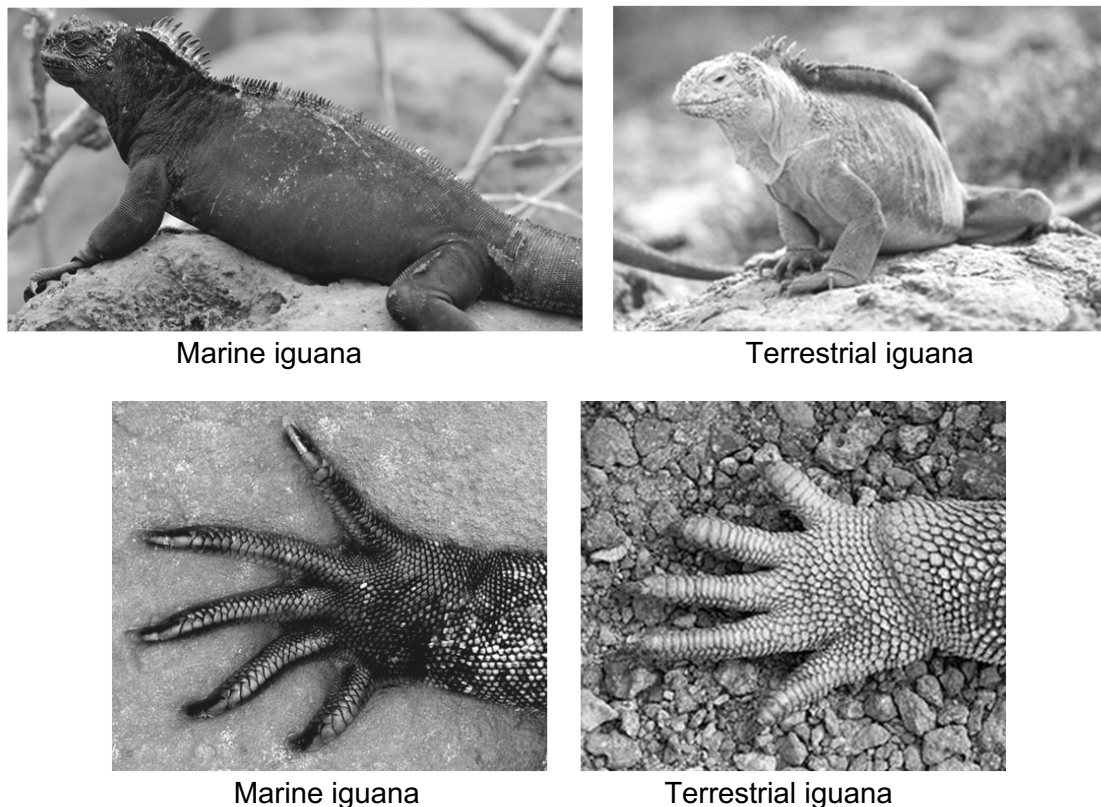


Fig 8.1

- (a) Based on the information provided, explain how marine iguanas are better adapted to the marine environment compared to terrestrial iguanas.

1. They have **darker skin tone** and **longer/sharper claws** (compared to the terrestrial iguana counterparts);
 2. Dark skin tones allow the iguanas to **rapidly absorb heat** (from sunlight) **after losing body heat to the cold waters**;
 3. Their claws are long / sharp for **clinging to rocks** on shore exposed to **strong waves**;
- [3]

[Turn over

Scientists have observed 17 populations of the Galapagos marine iguanas, all belonging to the same species, on the different islands in the archipelago.

The *cytochrome b* (*cytb*) gene of the mitochondrial genome was analysed and compared across the 17 iguana populations. Based on the analysis, scientists were able to identify three major clades as shown in the phylogenetic diagram in Fig. 8.2.

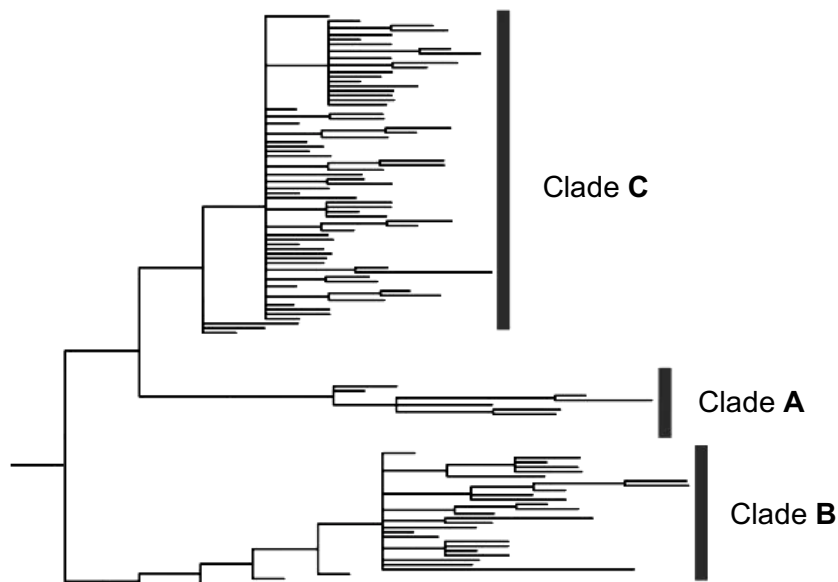


Fig. 8.2

- (b) (i) Define what is meant by a “clade” as shown in Fig. 8.2.
A group consisting of a single common ancestor and all its descendants, (where all members of the clade share similarities in DNA sequences of the *cytb* gene);
R! ref. to species
 [1]
- (ii) Explain what the branch lengths of the phylogenetic diagram in Fig. 8.2 represent.
 1. **Branch lengths show the number of differences in base sequence of DNA between descendants;**
 2. **Which arose due to mutations accumulated since divergence from the last common ancestor;**
 [2]

Fig. 8.3 maps the phylogeographic distribution of the three clades on the Galapagos islands, where the circles represent the proportion of the different clades of marine iguanas on each island.

It is known that the volcanic islands of Galapagos did not emerge at the same time, but progressively over a period of three million years. The map indicates the age of each island in mya (million years ago). The nearest mainland country, Ecuador, is 900 km east of these islands.

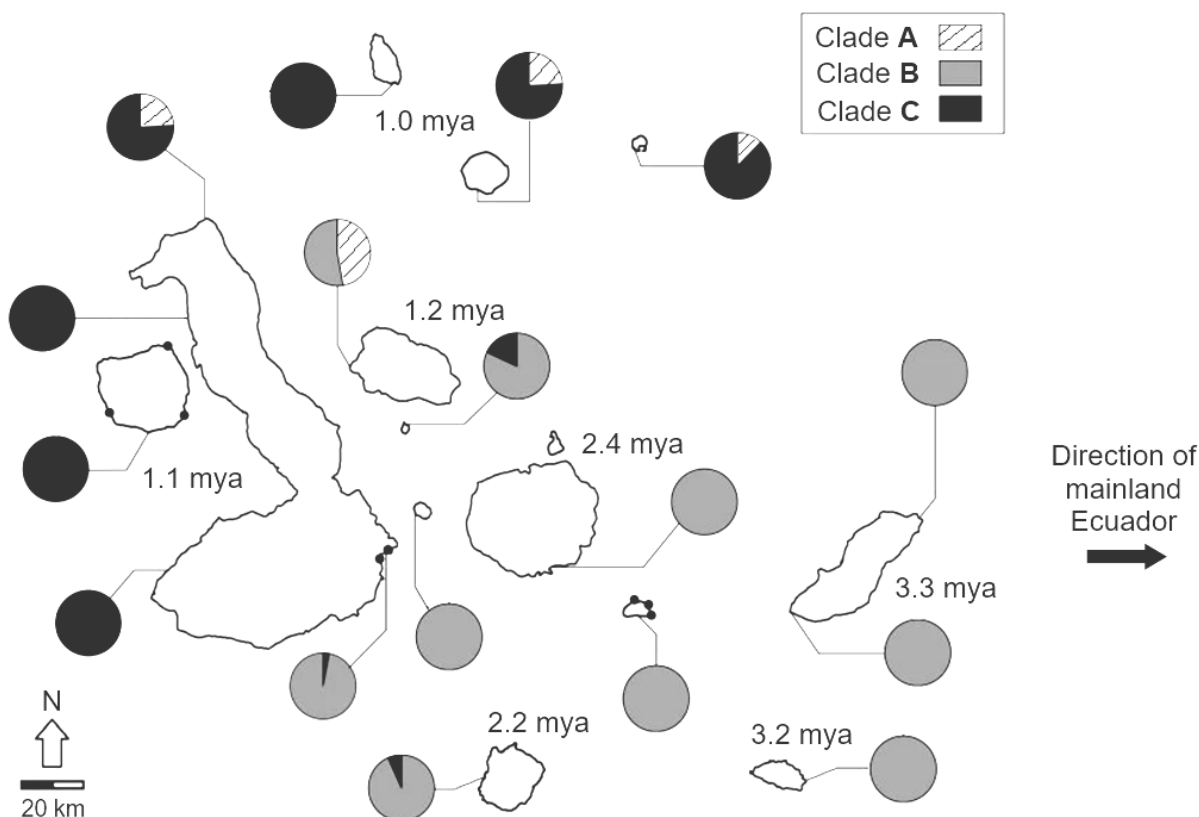


Fig. 8.3

- (c) (i) With reference to Fig. 8.3, describe how the phylogeographic distribution of marine iguanas (*A. cristatus*) belonging to clades B and C differ.

Iguanas belonging to clade B are found predominantly on the **islands in the East and South** while iguanas belonging to clade C mainly inhabit the **islands in the West and North** / North-West;

..... [1]

[Turn over

(ii) Suggest an explanation for the difference in phylogeographic distribution described in (c)(i).

1. The **Eastern islands formed earlier** between **3.2 to 3.3 mya**, followed by the **West /Northern islands** which are formed between **1.0 to 1.2 mya**;
2. The **ancestral** population could have **colonised the Eastern islands from mainland Ecuador**;
3. **New clades may be formed** due to geographical isolation by the **water bodies**, which **limited gene flow** between populations on different islands;
4. **Genetically similar/related populations** tend to be found on islands in **geographical proximity**;
5. **Different islands** may have **varying environmental conditions** which exert different selection pressures;
6. Hence **different clades** may experience **different reproductive success** on each island, resulting in the different proportions of iguanas from each clade;

max 3m

..... [3]

[Total: 10]

- 9 Glucose is an important substrate in cellular respiration. Fig. 9.1 shows the mTOR signalling pathway that drives cell growth by greatly stimulating glucose uptake and utilisation in a normal cell.

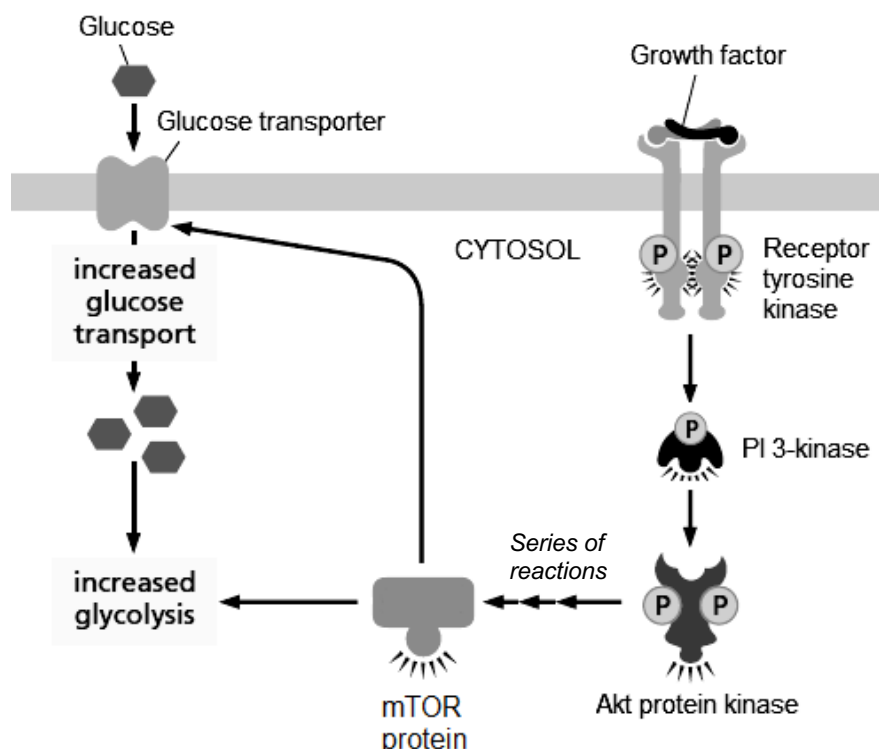


Fig. 9.1

- (a) (i) State one chemical property of the growth factor.
It is hydrophilic / polar / charged;
 [1]
- (ii) Describe how the binding of PI 3-kinase to the receptor tyrosine kinase leads to the cellular responses in Fig. 9.1.
1. PI 3-kinase binds to specific phosphorylated tyrosine residues on the receptor and becomes phosphorylated and activated;
 2. PI 3-kinase binds to Akt protein kinase, phosphorylating and activating it, which in turn activates other protein kinases via a phosphorylation cascade;
 3. Last protein kinase / relay protein in the phosphorylation cascade activates mTOR protein, increasing glucose transport / glycolysis;
 4. This is due to translocation of glucose transporters to the cell surface membrane, increasing permeability of cell to glucose / activation of enzymes which carry out glycolysis;

max 3m

..... [3]

[Turn over

The abnormal growth of cancer cells is highly dependent on increased glucose uptake and reliance on glycolysis for ATP production.

Recently, it has been suggested that having a ketogenic diet may be more inexpensive and easier to adopt than certain traditional anti-cancer therapies. A ketogenic diet is characterised by a high-fat and low-carbohydrate intake which is expected to decrease blood glucose levels and conversely increase ketone body levels. It is found that ketone bodies cannot be utilised by cancer cells.

Fig. 9.2 shows the effect of a ketogenic diet on a normal cell, where the decrease in glucose availability prevents some cellular processes from occurring.

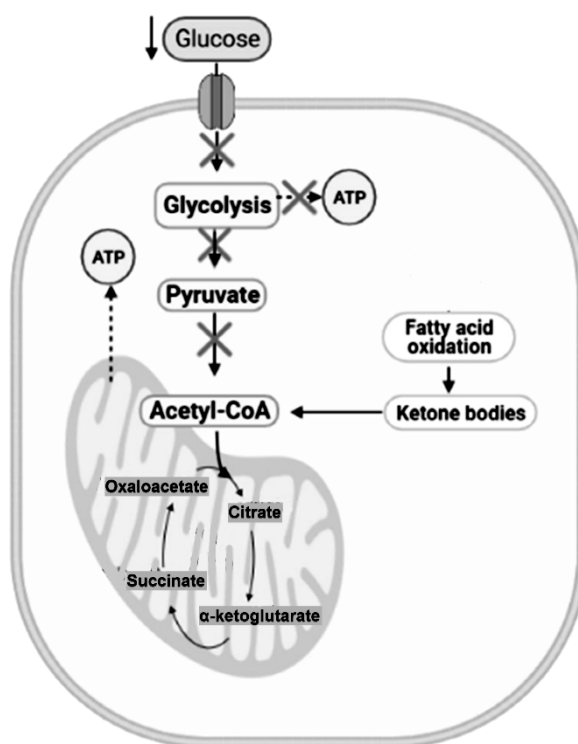


Fig. 9.2

- (b) (i) Describe how a ketogenic diet could act as a possible anti-cancer treatment.
1. Cancer cells **would not be able to synthesise acetyl-CoA** due to **lack of glucose** and **inability to utilise ketone bodies**;
 2. **ATP cannot be produced** by substrate level phosphorylation during glycolysis, which **cancer cells are reliant on**;
 3. **ATP cannot be produced** by substrate level phosphorylation during Krebs cycle (as acetyl-CoA is not produced);
 4. **ATP cannot be produced** by oxidative phosphorylation due to the **lack of NADH and FADH₂** molecules (as Krebs cycle does not occur);
 5. **No ATP is available** for the **growth and proliferation of cancer cells**;

max 4m

..... [4]

- (ii) While ketogenic diets show promise in limiting cancer cell proliferation, there are also concerns regarding its usage over a longer period of time.

Suggest the possible implications of having a ketogenic diet in the long term.

1. There is a need to evaluate effect of different fatty acids as **certain fats are not healthy**;
2. A diet high in **fat** may lead to other **health complications** such as obesity;
3. **Nutrient deficiencies** could develop depending on what kinds of food are part of the diet (e.g. consuming less staple food such as rice and noodles);
4. **Not all cell types in the body may be able to utilise fatty acids** as a source of energy;

max 2m

..... [2]

[Total: 10]

[Turn over

10 One distinctive feature of the adaptive immune system is the ability to generate millions of different antibody molecules to specifically target different foreign antigens.

- (a)** Fig. 10.1 shows the structure of a typical antibody, where two major segments of the antibody have been labelled with blank boxes.

Using the letters '**S**' to represent somatic recombination, '**H**' to represent hyper-mutation, and '**C**' to represent class switching, indicate in the blank boxes which segment of the antibody is affected by each of the three processes (each box may be filled by more than one symbol).

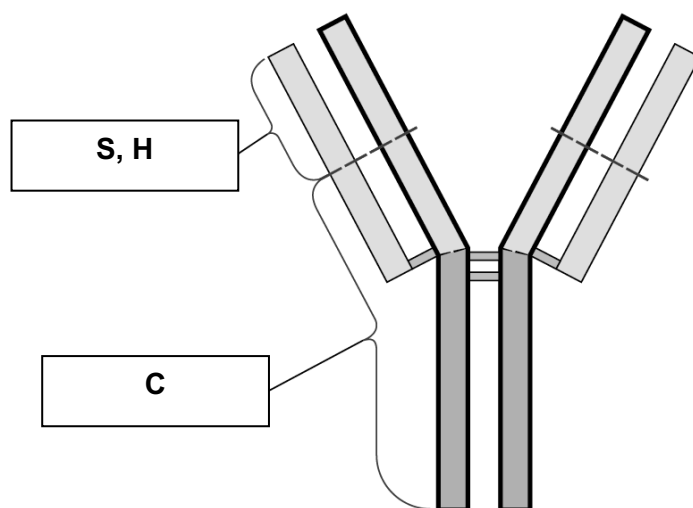


Fig. 10.1

[3]

Fig. 10.2 shows the concentrations of different antibody types following an initial exposure to a specific antigen and a repeated exposure to the same antigen.

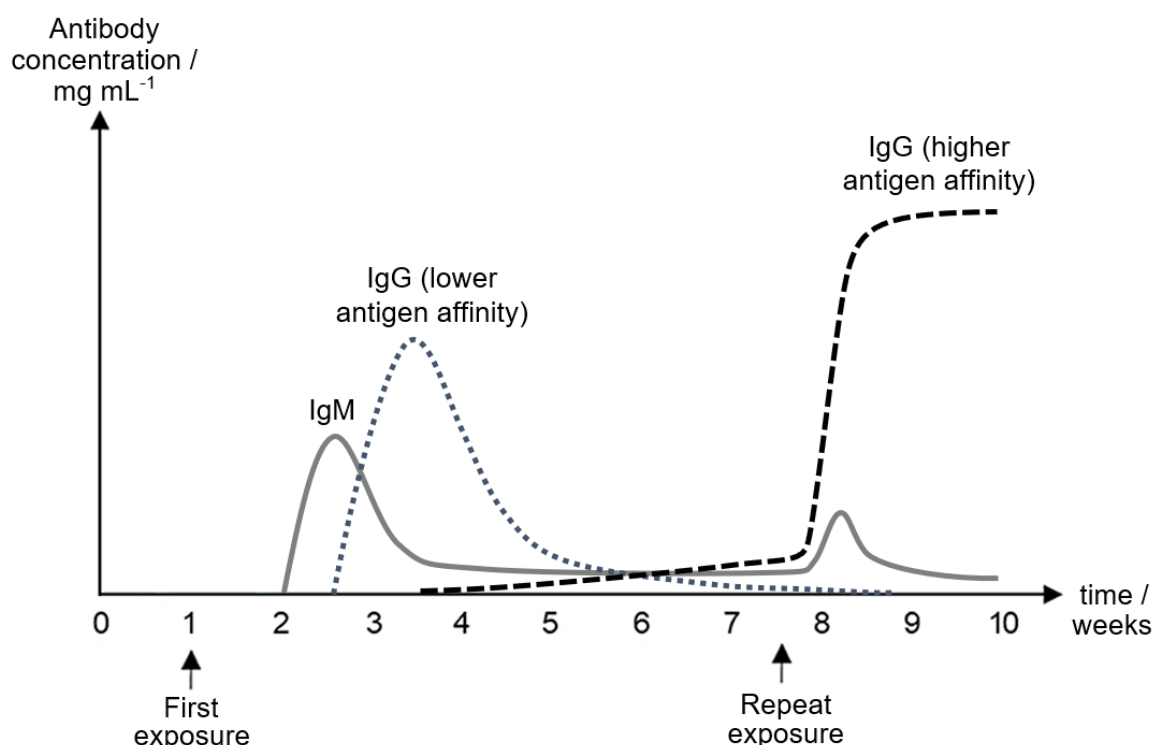


Fig. 10.2

(b) With reference to Fig. 10.2, explain how the type of antibody secreted by B cells changes during the primary immune response.

1. **IgM** is the **first class of antibody to be secreted** by B cells, with secretion **starting at week 2 / peaking at week 2.5**;
2. **IgG** secretion occurs **later** only **after class switching** has taken place, with secretion **starting at week 2.5 / peaking at week 3.5**;
3. **IgG** of **higher binding affinity to antigen** is only secreted **after somatic hyper-mutation / affinity maturation** has taken place, with a low concentration secreted **starting at week 3.5** during the primary immune response;

..... [3]

[Total: 6]

[Turn over

- 11 Biodiversity refers to the wide variety of species found in various habitats. To reduce the loss of global biodiversity, large areas of habitat have been protected.

Fig. 11.1 shows the changes in global biodiversity and in the total area protected from 1965 to 2005, in terrestrial and marine habitats.

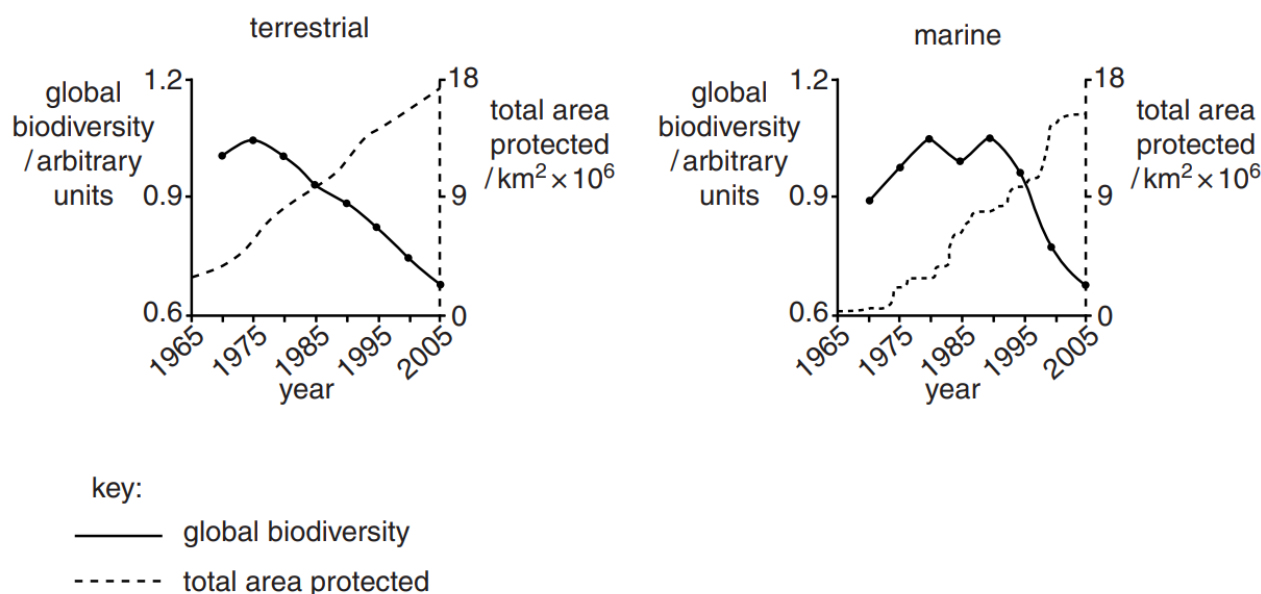


Fig. 11.1

- (a) (i) Compare the relationship between total area protected and global biodiversity in marine habitats:
- between 1970 and 1990, and
 - between 1990 and 2005.

1. Between 1970 and 1990, as total protected areas increased, global biodiversity increased;
2. As total protected areas increased from $0 \text{ km}^2 \times 10^6$ in 1970 to $8 \text{ km}^2 \times 10^6$ in 1990, global biodiversity increased from 0.9 arbitrary units to 1.05 arbitrary units;
3. exceptions; e.g. dip from 1980 till 1985;
4. Between 1990 and 2005, as total area protected increases, global biodiversity decreases;
5. As total protected areas increased from $8 \text{ km}^2 \times 10^6$ in 1990 to $16 \text{ km}^2 \times 10^6$ in 1990, global biodiversity decreased from 1.05 arbitrary units to 0.7 arbitrary units;

A! a.u. after defining once
max 4m

..... [4]

(ii) Suggest why a smaller area of marine habitats has been protected than of terrestrial habitats.

1. **marine environments are difficult to patrol/ monitor;**
2. **lack of public, awareness / interest;**
3. **International ownership issues; A! example**
4. **difficult to, set /mark / recognise, boundaries;**
5. **AVP; e.g. problem of mobile populations thus difficult to decide on area to protect;**

max 1m

..... [1]

[Total: 5]

[Turn over